

Machine Learning

FAKE NEWS DETECTION USING NLP, TF-IDF AND BERT

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PROBLEM STATEMENT

- Fake news spreads rapidly through online platforms.
- Misleading information can influence people's opinions and decisions.
- Fake and real news can have very similar writing styles.
- Manual fact-checking is slow and difficult for large amounts of news.
- An automated system is needed to identify potentially fake news.



MOTIVATION

Why This Project?

- Reduce the spread of misleading information.
- Automatically analyze large amounts of news.
- Detect suspicious textual patterns.
- Reduce the effort required for manual checking.
- Compare traditional ML with advanced NLP models.
- Support faster identification of potentially fake news.



LITERATURE REVIEW

25 research papers were studied.

Main approaches found:

- N-Gram + Machine Learning
- SVM
- Logistic Regression
- Decision Trees
- Random Forest
- RNN /CNN / LSTM
- BERT
- FakeBERT
- Multimodal models
- Hybrid models
- Explainable AI

Common limitations:

- Dataset bias
- Limited datasets
- High computational requirements
- Dependence on social-media information
- Text + image requirements
- Difficulty in real-time implementation
- Limited generalization

RESEARCH GAP



From the literature, we observed:

- Traditional ML methods are simple but may have limited contextual understanding.
- Deep learning models require more computational resources.
- Some approaches depend on social-media information.
- Multimodal models require additional image data.
- Dataset quality and bias can affect performance.
- BERT-based models provide better contextual understanding but are more resource-intensive.

DATASET

Dataset: Fake News Dataset

- Total Records: 2,000
 - Total Features/Columns: 4
 - Target Classes: 2
 - Fake News
 - Real News
- Fake News: 1,000 records
 - Real News: 1,000 records
 - Dataset is balanced: 50% Fake + 50% Real
 - Missing Values: 0
 - Duplicate Records: 0

COLUMN

serial_number

title

text

label

DESCRIPTION

Unique record number

Headline of the news

Full news article/content

Fake or Real

DATA PREPROCESSING

Our Python work performs:

1. Load dataset
2. Check missing values
3. Check duplicate records
4. Separate input and target
5. Analyze numerical and categorical features
6. Clean news text
7. Prepare text for NLP processing
8. Split data for model training and testing

Dataset Quality

- Missing values: 0
- Duplicate records: 0
- Balanced classes: 1000 Fake / 1000 Real



PROPOSED SYSTEM ARCHITECTURE

System Workflow

Fake News Dataset



Data Preprocessing



NLP Text Processing



Feature Extraction

TF-IDF



Logistic Regression



Fake / Real Prediction

Advanced Pipeline

News Text → BERT → Fake / Real

Final Output:

FAKE NEWS / REAL NEWS

MACHINE LEARNING MODELS

1. Logistic Regression

- Supervised classification algorithm.
- Simple and fast.
- Works well with TF-IDF features.
- Suitable for binary classification.

TF-IDF → Logistic Regression → Prediction

2. BERT

- Transformer-based NLP model.
- Understands contextual relationships.
- Handles complex language patterns.
- Used as the advanced comparison model.

BERT → Contextual Representation → Prediction



FUTURE SCOPE

- Use larger and more diverse datasets.
- Add real-time fake news detection.
- Develop a web/mobile application.
- Include image and social-media information.
- Improve BERT fine-tuning.
- Add explainable AI to show why news is classified as fake.

CONCLUSION

- Fake news is a major problem in digital communication.
- NLP can automatically analyze news content.
- TF-IDF + Logistic Regression provides a simple ML approach.
- BERT provides deeper contextual analysis.
- Comparing both models helps understand their strengths and limitations.
- The system can act as a supportive tool for identifying potentially misleading news.



THANK YOU